

1. Set

$$g(x, y) = \begin{cases} \frac{x^2 y^2}{x^4 + y^4}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Show that $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$ both exist at $(0, 0)$. What are their values at $(0, 0)$?

(b) Show that $\lim_{(x,y) \rightarrow (0,0)} g(x, y)$ does not exist.

2. Set

$$f(x, y) = \frac{2x^2 y}{x^4 + y^2}.$$

Evaluate the limit of $f(x, y)$ as (x, y) approaches the origin along

(a) the x -axis;

(c) the line $y = mx$;

(b) the y -axis;

(d) the parabola $y = ax^2$.

Does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?

3. Find the gradient at P .

(a) $f(x, y) = \ln(x^2 + y^2)$; $P(2, 1)$.

(b) $f(x, y) = \frac{2x}{x-y}$; $P(3, 1)$.

4. Find the directional derivative at the point P in the direction indicated.

(a) $f(x, y) = xe^y - ye^x$ at $P(1, 0)$ in the direction of $3\mathbf{i} + 4\mathbf{j}$.

(b) $f(x, y) = \frac{2x}{x-y}$ at $P(1, 0)$ in the direction of $\mathbf{i} - \sqrt{3}\mathbf{j}$.

5. Find the unit vector in the direction in which f increases most rapidly at P and give the rate of change of f in that direction; find the unit vector in the direction in which f decreases most rapidly at P and give the rate of change of f in that direction.

(a) $f(x, y) = y^2 e^{2x}$; $P(0, 1)$.

(b) $f(x, y) = x + \sin(x + 2y)$; $P(0, 0)$.